Related Work & Research Gaps

System	Venue	Multi-agent Router	VLM Disease	Soil Sensor	Weather Prediction	XAI Farmer UI	Voice / Indic	Market Price	Deployed
Farmer.Chat [?]	ACM 2024	×	× (Generic)	×	✓ API	×	✓ Partial	×	✓ 565k users
AgroAskAI [?]	arXiv 2024	✓ Chain	×	×	✓ Dedicated	×	✓ Partial	×	×
AgriGPT [?]	arXiv 2025	✓ Data engine	✓ VL ext.	×	✓ Partial	×	×	×	×
AgriM-LLM [?]	Preprints 2026	×	✓ Best VE	×	×	×	×	×	×
AgriDoctor (Base) [?]	IEEE 2025	✓ Router+5	✓ Strong	×	×	×	✓ CN/EN only	×	×
AgriBuddy [?]	BRAC 2025	✓ Partial	✓ CNN rice	×	✓ Partial	×	×	×	✓ Partial
PhenoAssistant [?]	Nature 2026	✓ Yes	✓ Research	×	×	×	×	×	× (Lab)
Agentic AI [?]	MDPI 2026	✓ 4 agents	✓ Rice CNN	✓ Partial	✓ Dedicated	×	×	×	×

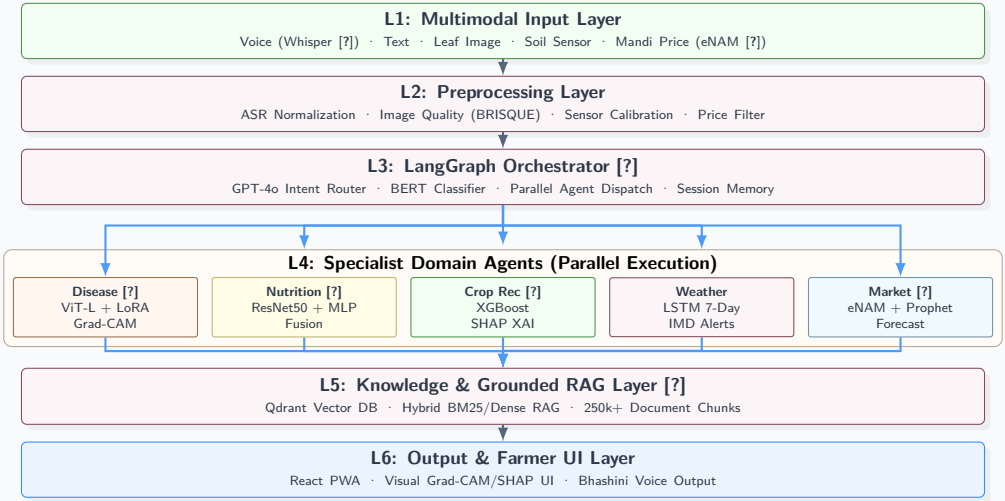
Key Research Gaps Identified

- **Soil Sensor Fusion** and **Market Price** features are completely missing from existing systems.
- No prior platform supports explainable AI (**XAI**) for farmer UIs or comprehensive **Indic speech** interfaces.

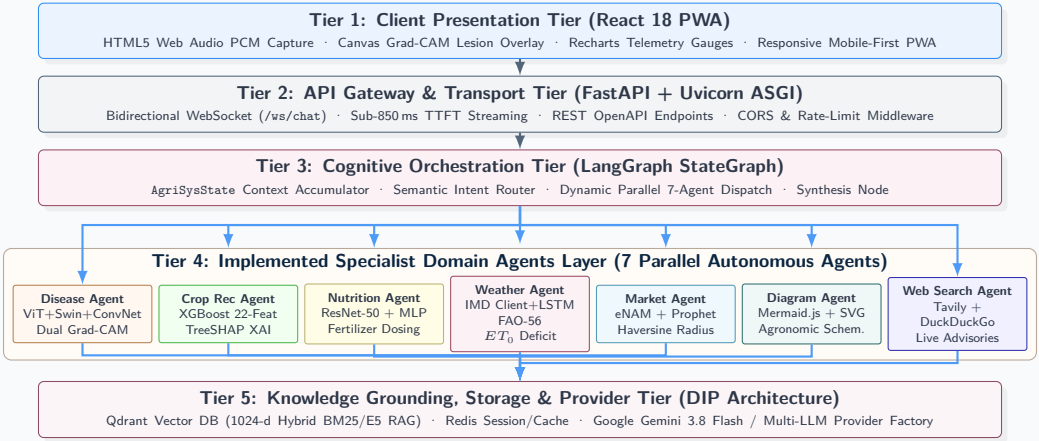
Problem Statement

Current agricultural AI systems are fatally fragmented: deployed chatbots lack visual diagnostics, advanced vision models lack interactive conversational interfaces, multi-agent setups ignore live market data, and none deliver explainable advice in native Indic languages. Consequently, no existing framework integrates speech, image, soil sensors, weather prediction and live market data into a single trustworthy session.

System Architecture



Implementation Architecture



Technical Implementation Details

Multi-Agent Orchestration & Streaming Architecture

- **LangGraph Multi-Agent DAG:** Asynchronous state machine coordinating query dispatch across 7 specialist agents with `AgriSysState` memory.
- **Real-Time WebSocket Protocol:** Bidirectional `/ws/chat` gateway delivering sub-850 ms Time-to-First-Token (TTFT) and delta streaming.
- **SOLID Provider Abstraction (DIP):** Strict decoupling via `BaseLLMProvider`; dynamically interfaces Gemini 3.8 Flash, Groq, and Ollama.
- **Dynamic Diagram Agent:** Automated generation of interactive Mermaid workflows (IPM trees, crop timelines) and 2D measured SVG schematics (spacing grids, drip lines).

Multimodal Machine Learning, XAI & Hybrid RAG

- **Crop Disease Diagnosis:** Accuracy-weighted tri-model ensemble (ViT-L/16, Swin, ConvNeXt) + BRISQUE QA + PyTorch Grad-CAM lesion heatmaps.
- **Crop Rec & Nutrition Fusion:** 22-feature XGBoost with TreeSHAP; ResNet-50 + IoT soil sensor (NPK, pH) fusion MLP for fertilizer dosing.
- **Market & Web Search Agents:** eNAM/Agmarknet scraper with 7-day Prophet forecast; live Tavily/DuckDuckGo search for real-time agricultural advisories.
- **Two-Stage Grounded Hybrid RAG:** Qdrant vector DB (1024-d Multilingual E5) + BM25Okapi sparse search merged via RRF ($k = 60$) and Cross-Encoder reranking.